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## Using dataExplorer as a Data Harmonization Platform: PCR-RFLP Genotyping Data

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**Goal of this exercise:** To explore how data harmonization and standardized metadata can enable efficient integration of newly published *Toxoplasma gondii* isolate data into an existing curated genotype database.

### Background

In 2014, Jitender Dubey, Chunlei Su, and colleagues [published a landmark synthesis](#) of global *Toxoplasma gondii* genotype data (SHWAB EK, ZHU X-Q, MAJUMDAR D, et al. *Geographical patterns of Toxoplasma gondii genetic diversity revealed by multilocus PCR-RFLP genotyping*. Parasitology. 2014;141(4):453-461).

The study compiled and standardized reported data for more than 1,400 *T. gondii* isolates collected worldwide through the end of 2012.

The dataset is based on **multilocus PCR-RFLP genotyping**, a standardized approach that uses a panel of genetic markers to characterize allelic patterns and assign isolates to distinct genotypes. Each record represents a single isolate and includes information such as geographic origin, host species, genotype assignment, and allele calls across the marker panel.

Metadata captured for each isolate includes collection location (country, continent, and coordinates where available), common and scientific host names, and a PMID linking each record to the original publication.

The harmonized genotype table was later incorporated into [a searchable ToxoDB resource](#), and additional isolate data from subsequent publications have been incorporated over time. We have downloaded the PCR-RFLP genotyping data from ToxoDB and have incorporated it into dataExplorer (see dataset: "**T. gondii PCR-RFLP genotypes in ToxoDB rel. 68**").

In this exercise, we will explore how data harmonization principles- shared variables, standardized metadata, and consistent data structures- can make it easier to identify,

extract, and integrate newly published isolate data into an existing reference dataset. The goal is to consider whether establishing and maintaining a consensus data model could accelerate future database updates and improve long-term interoperability.

## Exercise

1. Explore the aggregated “**T. gondii PCR-RFLP genotypes in ToxoDB rel. 68**” dataset on dataExplorer
  - Find the dataset and open an analysis. Review the list of variables
    - If you were interested in learning more about genotypes of chickens in Brazil, how would you find the relevant publications?
  - Explore the data in the interactive map
2. Finding additional data to integrate: This dataset would benefit from regular updates as toxo researchers are generating additional PCR-RFLP data all the time. We looked for recent papers which published PCR-RFLP data from various species in different parts of the world and found **3 example datasets** where PCR RFLP genotyping data is displayed and available in tables that would contribute to and enhance this dataset.
  - Chen YJ et al. *Dolphins share Toxoplasma gondii Type II genotypes with terrestrial animals: Evidence of terrestrial T. gondii contamination in marine environments*. Vet Parasitol. 2025 Apr;335:110439. doi: 10.1016/j.vetpar.2025.110439. **PMID: 40048863**.
    - See Table 3
  - Valenzuela-Moreno LF et al. *New genetic variants of Toxoplasma gondii isolates and a novel ROP5 allele in free-range chickens from Tabasco, México*. Parasitology. 2025;152(10):1083-1091. doi:10.1017/S0031182025100401. **PMID:40588766**.
    - See Table 1
  - Remor-Sebolt, AP et al. *Isolation and molecular characterization of Toxoplasma gondii from naturally infected seabirds on the southern coast of Santa Catarina state, Brazil*. Parasitol Res 124, 113 (2025). doi: 10.1007/s00436-025-08558-1. **PMID: 41081955**.
    - See Table 2
3. Integration of additional data - Step 1: Review the raw data in these 3 datasets. The PCR-RFLP genotype tables from the three publications were downloaded and combined into a single dataset – “**T. gondii PCR-RFLP genotypes in sea birds (Brazil), chickens (Mexico), and dolphins (Australia) - RAW**” – without any harmonization or standardization of variables or values.

Columns were matched solely based on exact column names; therefore, variables representing the same concept but labeled differently across studies were retained as separate columns in the combined dataset.

When columns shared the same name, their records were merged into a single variable field regardless of differences in value coding or terminology. As a result,

values that represent the same category but were recorded using different labels (e.g., alternative abbreviations, spellings, or nomenclature) remain distinct and are not automatically grouped or reconciled.

The combined “**RAW**” dataset should therefore be considered a direct aggregation of the source tables rather than a harmonized cross-study resource.

Explore “**T. gondii PCR-RFLP genotypes in sea birds (Brazil), chickens (Mexico), and dolphins (Australia) - RAW**” on dataExplorer.

- Take a look at the 3 different variables are used to identify isolates or samples:
  1. *Isolate*
    - Which dataset used this identifier?\_\_\_\_\_
  2. Bioassay Number
    - Why would we say that this identifier is represented in a format that is not “tidy data”?\_\_\_\_\_
    - Tidy data* is a standardized way of organizing data in which:
      - Each **row** represents a single observation (record).
      - Each **column** represents a single variable (attribute).
      - Each **cell** contains a single value.
      - Each type of observational unit is stored in its own table.
      - Note how this identifier combines isolate ID and bioassay number into a single variable
  3. *T. gondii*
    - What is an issue with the variable name for this identifier?  
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What differences do you notice between this dataset and “**T. gondii PCR-RFLP genotypes in ToxoDB rel. 68**”? Most striking issues across the 3 additional small datasets:

- **Metadata** is not present in tables in the publications and will have to be manually added from the paper text: country, GIS coordinates, host species, PMID.
- **Names of columns:**
  - The same 11 core markers appear in all datasets but are named inconsistently.
  - The ToxoDB Genotype variable also appears in all 3 datasets but is named inconsistently (see the variables labeled: T. gondii, ToxoDB PCR RFLP genotype, and Genotype).
  - Every dataset has an isolate/sample identifier, but the column is named differently in each
  - The SAG2 column name is not harmonized at all: the Brazilian sea birds and Mexico chickens datasets use different names for this variable, and the Australia dolphins dataset splits it into two separate columns (5' and 3').
- **Marker set:** Australia dolphins dataset has an extra marker (B1) not present in any other dataset- a decision point about whether to expand the schema.

- **ToxoDB genotype format** (see the variables labeled: T. gondii, ToxoDB PCR RFLP genotype, and Genotype):
    - every small dataset uses #N (e.g., #36), while the main dataset uses plain numbers.
    - Novel genotypes get free-text annotations like NEW #344 or #348, this study, which breaks any automated join.
  - **Allele notation for ambiguous calls:**
    - All datasets use Roman numerals (I, II, III) for alleles, but non-canonical values differ: for example, nd and ND. for not determined
    - IIorIII, II/III, II or III, and II / III are all in use across datasets for the same thing.
  - Datasets often contain **additional columns** that the main dataset does not have and that could enrich it, such as tissue of isolation and status of host-stranded/ wild/ captive/ food animal/ domestic...
4. Integration of additional data - Step 2: VEuPathDB staff harmonized PCR-RFLP genotype tables in these 3 datasets by ensuring standardized variable labels and values were used. Metadata such as country, GIS coordinates, host species, PMID were also added. Explore the resulting dataset: **“T. gondii PCR-RFLP genotypes in sea birds (Brazil), chickens (Mexico), and dolphins (Australia) - HARMONIZED”**
- How does the total number of variables differ between the raw and harmonized datasets?\_\_\_\_\_
  - Why were the 5'-SAG2 and 3'-SAG2 variables not included in the harmonized dataset? \_\_\_\_\_

What are some positive effects of harmonizing these 3 datasets?

- Improved comparability: Harmonization standardizes variable names, definitions, coding schemes, and units, allowing equivalent measurements from different studies to be analyzed together. This reduces ambiguity and makes cross-study comparisons more meaningful.
- Increased sample size, global coverage, and statistical power: By combining data from multiple sources into a common structure, researchers can analyze larger numbers of observations than would be possible within any single study. This can improve the precision of estimates and increase the ability to detect patterns or associations.
- Enhanced data discoverability and reuse: Standardized variables and metadata make datasets easier for other researchers to understand, search, and reuse, supporting FAIR data principles.
- Reduced data preparation effort: Researchers can spend less time cleaning, recoding, and mapping variables before analysis because much of this work has already been performed during harmonization.
- Cross-study and meta-analytic analyses: Harmonized datasets enable analyses that span geographic regions, time periods, host species, or study designs, facilitating broader scientific questions that cannot be addressed using individual studies alone.

- Improved data quality: The harmonization process often identifies inconsistencies, errors, duplicate records, missing metadata, or incompatible coding schemes, leading to higher-quality datasets.
- Reproducibility and transparency: A documented harmonization process provides a clear record of how variables were mapped and transformed, making analyses easier to reproduce and interpret.

What are some negative effects of harmonizing these 3 datasets?

- Loss of data richness
  - Mexico (lost municipality, tissue type, tachyzoite density variables)
  - Australia (lost PCR-RFLP data for the B1 locus)

5. Integration of Additional Data - Step 3: Because the datasets have been standardized to a common format, their records can be integrated into the aggregated dataset, expanding its coverage and increasing the number of available genotyped isolates.

The harmonized data from these three datasets has been combined into the aggregated dataset and saved as **"RFLP\_genotypes\_ToxoDB\_rel68\_plus3.txt"**

You will now upload the **"EXPANDED, AGGREGATE"** dataset into dataExplorer.

- Download the **"ToxoDB\_rel68\_plus3"** zip folder, which contains the combined data file and associated variable attributes.
- Click "Upload my Dataset", located under the Workspace dropdown menu in dataExplorer's header. Name your dataset **"EXPANDED, AGGREGATE"**, fill out a brief Summary, and attach the data file and attributes file in the correct locations. Now upload the dataset!
  - Data file: RFLP\_genotypes\_ToxoDB\_rel68\_plus3.txt
  - Attributes file: RFLP\_genotypes\_toxoDB\_rel68\_plus\_3\_var\_annot.txt
- How many records does your **"EXPANDED, AGGREGATE"** dataset contain? \_\_\_\_\_
- Does this dataset contain the number of records you expect? *Hint: add up the number of records in "T. gondii PCR-RFLP genotypes in ToxoDB rel. 68" and "HARMONIZED"* \_\_\_\_\_
- Let's explore why you are seeing fewer records in the **"EXPANDED, AGGREGATE"** than you would expect:
  - How many isolates from the Australia dolphins manuscript were included in the **"HARMONIZED"** dataset? \_\_\_\_\_
  - How many isolates from the Australia dolphins manuscript were included in the **"EXPANDED, AGGREGATE"** dataset? *Hint: find PMID:40048863* \_\_\_\_\_
  - Why do you think that fewer isolates were included in the harmonized dataset than were included in the original manuscript? (*Hint: look at Table 3 in DOI: [10.1016/j.vetpar.2025.110439](https://doi.org/10.1016/j.vetpar.2025.110439)*)

## Discussion

- Community standards
- International database
- Core group of curators

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